



PREDICTING THE NBA ALL-STAR ROSTER THROUGH MACHINE LEARNING

REVEALING HIDDEN BIAS IN NBA ALL-STAR
PICKS

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Abstract

In the last few decades, the professionalism in the sports environment had sky rocketed in both aspects competence and financially wise, which may explain why more technical teams using methods like machine learning and artificial intelligence commence to play a role in the field as well. Given an NBA data set of performance of players during the last three decades and extracted information about NBA All-Star rosters for the exact period, it was predicted with a 0.93 accuracy, 0.68 true positives ratio and 0.98 true negatives ratio and a macro average of 0.83 the prediction of NBA All-Star players. Certain features like "name" and "country" were eliminated to avoid bias in the learning process. Moreover, chi-square statistical tests were performed in different categories of the true positive instances of the random forest model, finding a significance p-value of 0.011 after controlling the true positives for the variable "college". The results showed that besides extracting some features considered as bias like "name" and "country", the weight gave to the variable "college" had a significant value in the mismatch of the algorithm when predicting the selected players, possibly translating into the people who chose those players who are the NBA fans, the specialized media and the NBA coaches assigning more relevance to the early career stages of players than it actually deserves. Aiming to unveil the bias in the sport and evidencing that there is a society expectation in players career success based in the chi-square results in the true positives data set of the random forest model for the All-Star roster selection.

1 DATA SOURCE, ETHICS, CODE, AND TECHNOLOGY STATEMENT

1.1 *Source/Code/Ethics/Technology Statement Example*

The programming language used for the writing of the master thesis was Python and the environment used was VS code. The data sets used for

the research were publicly available online, one part was extracted from the website Kaggle, whereas the rest of it was mined by the researcher for this experiment from a historical NBA statistics website, mined in an Excel worksheet and then imported to python VS code environment. The elaboration of the writing was enhanced by the use of Thesaurus.com, Grammarly, and Microsoft word spelling corrector. The main libraries used in VS code environment were “sklearn”, “matplotlib”, “pandas” and “scipy.stats” for the Machine Learning, the plotting, the data framing, and the statistical models, respectively. The figures that were not represented by the “matplotlib” package in python were elaborated by the author using Microsoft word.

2 INTRODUCTION

Since I have use of memory, I can remember myself attached to a tennis ball, football, basketball, or some kind of sport element in some different shape or discipline. Sports, like Basketball, is not only a benefactor for physical and mental health but also one of the greatest sources of entertainment around the world. Every country in the world practice different sports disciplines, and most of the times also in different division levels. This different divisions and disciplines have developed so much due to technology and professional level of competition, that the budget for these organizations surpass the billions of euros annually. Besides the growth of their expending, other shareholders surrounding Basketball are also eager to this particular topic of identifying the top players in National Basketball Association (hereafter; NBA) and this are the Club Management and more precisely professional players scouting.

On the other hand, most societies as well as competing in sports, generate a lot of expectation when wondering who the next great athlete or athletes are going to be. According to this general thinking regarding sport competitors is that the research approaches a method to analyze if, in the case of NBA All-Star players, are there any permeable categories that might be in the expectation process, given the historical NBA players from seasons in between 1996 to 2022 and the ones selected from fans, specialized media and team coaches to get selected in a prestigious list of players of 24 athletes per season.

Club management may face challenges in the decision making of spending their budget on the future of the club(Guevara et al., 2021). A prediction model with a high performance on identifying the top and quality players can neutralize these challenges, helping reduce the financial risk and help these institutions be more financially sustainable in the long run. Professional scouts basketball wise, is related to the professional that works

on discovering new talent, or identifying strategic players that may adjust to different teams. On another order of relevance, the conclusions of this research is useful for this profession too, as it reveals the different features of basketball players and relates it to a certain quality or a certain bias that may lead scouts to hire certain players for certain teams as well as discovering talent for specific projects more efficiently.

The goal of this research is to predict by the use of ML models the selection of NBA players for the NBA All-Star game. To compare the performance of this models and to check if there is a significant bias in some of the category features.

In order to do this, statistical chi-square tests were performed in the different category features in the best performing ML model. In the best model, the with lower performance was utilized, as well as some features with heavy bias influence such as "name" and "country" were eliminated.

Moreover, the gap to be filled by this study is in conscious in the decision making bias that may exist for the selection of this best-of-season players. As there are academic research's about 'decision game bias' (more about this under literature review section) for attributes like race or ethnicity. Meanwhile, there are no studies conducted about NBA All-Star selection bias regarding this selected group of NBA players, which lies on fan ballot voting, NBA coaches election, and specialized media votes.

Through this research, NBA directories and scouts are going to be able to identify which players are prone to the selection of the All-Star game using different binary classification Machine Learning (hereafter; ML) models with a more conscious approach as some influencing variables may be detected.

They will be capable of avoiding certain bias in the modelling in the sense that big potential in certain players might be overlooked due to some feature involving players. This research will be done comparing different binary classification models, like logistic regression, decision trees and random forest. Moreover, to what extent was a bias identified and may be interfering in the recruitment of the All-Star players. (Langel & Tillé, 2011).

As a result of this, is that we formulate the following three research questions:

RQ1 *To what extent can the selection of NBA All-Star players be predicted using a machine learning model?*

RQ2 *To what extent can a categorical feature bias the prediction outcome of NBA All-Star players?*

RQ3 *How do this different machine learning models perform compared to each other given the NBA dataset?*

3 LITERATURE REVIEW

So far, there are many studies that show that it is possible to use Machine Learning models to predict the all-star rooters for NBA. Furthermore, other type of bias such as racial biases and socioeconomic background have been demonstrated by previous researchers. This background research gives confidence that it is possible to investigate the categorical biases in this field, and this is the research gap the thesis research aimed for, understanding society in sports by implementing Machine Learning and statistical testing. Finally, it can be concluded that determining bias is a relevant topic in the use of data science in society that can be applied to NBA, other sports, and many other fields.

The NBA was founded in the year 1949, but it was not until 1984 when it was relaunched as an international entertainment endeavor. Concurrently, the first NBA All-Star game happened in the year 1951 (Grossman, 2014). It is a friendly and leisure match played by an East Conference recruited team against a West Conference recruited team from the NBA. These All-Star players are chosen via fan ballot and votes among the head coaches of each squad from each conference. There is a total of 24 players in the mentioned game, 5 starters on each side plus 7 substitutes.

Besides popularity and professionalism, the NBA is also an institution that has been working longstanding with data collection, thanks to that we can find many other researchers to support our investigation. Beginning from the research about Predicting all star player using a random forest model, it was found that the most relevant feature from a dataset from 1937 to 2011 was the number of points contributions made by minute of game play, a 0.925 accuracy was reached using this feature in a random forest model. (Soliman et al., 2017)

In accordance with the bias topic, another academic paper suggests that there is a racial discrimination from coaches towards players from a different race although they show a same level performance than other teammates from the same race as their respective coach. It also suggests that these bias decreases as they spend more seasons sharing the same team, making the conclusion that this discrimination is not towards individuals but a bias towards a group. (Zhang, 2017)

On the other hand, another research called Machine Learning on NBA Data Sets found that 'elementary statistics' are outperformed by the 'advanced' ones. Elementary statistics refer just to the collection of individual performances of players during games and not applying any further anal-

ysis to these values. Whereas the advanced ones are given due to the process of the data of these elementary features using ML models. For models ran on Decision Trees, the advanced statistics outperforms the elementary statistics significantly from 0.75 in elementary dataset to 0.91 in the advanced dataset. For a Random Forest of elementary statistics, the score was of 0.85 and of 0.88 in the advanced one. For a Gradient Boosting model elementary performed with an 0.82 whereas advances statistics outperformed with a 0.92 accuracy. (Wang & Fan, 2021)

Another NBA paper about bias and racism, suggests the following idea, similar to a previous statement but adding that these discrimination bias can be exacerbated under contexts of pressure when doing changes during a game. Meaning that the bias can be stronger during the flow of the game, and minor when selecting the starting team. However, it was shown that bias was still present when making deliberated decisions like choosing the starting team. (Schroffel & Magee, 2012)

An article regarding bias in another popular sport like soccer, suggested that the players with a higher socioeconomic background have better chances on succeeding, as well as education and performance, education and promotions, socioeconomic background and performance, socioeconomic background and promotions (Gandelman, 2009). Some of this concepts can trigger findings or helps us make conclusions when working on if there is a bias of this team selection.

Another research that worked with open-source statistics of the NBA, found that a hybrid model was the best performer to identify true positives in players selection for All-Star teams, as well as identifying True negatives for players who were not selected for the All-Star team. This hybrid model was composed by an Artificial Neural Network (hereafter; ANN) with just one hidden layer, and where the input neurons are the outputs of other models. The research mostly focused on the specificity and sensitivity, naming how precisely it could identify true positives and the true negatives in a test set. (Wang & Fan, 2021)

Moreover, another study suggests more characteristics for players to be considered by fans as an All-Star player, besides the sensitivity of win shares mentioned previously, they found interesting features in three-point shooters, making this characteristic a player more valuable. Furthermore, the 'masculine attitude' strictly related to the number of rebounds and free shots made per game, this was considered a valuable feature making a player more eager to be voted as a candidate for the All-Star team. (Levine, 2019)

Another investigation more related to the university sports field in the United States, suggested that university teams that have a longer winning streak in their seasons consecutively get a lower acceptance rate, receive

more donations, as well as academic applications and even making noticing an increment in the academic reputation (Stinson & Howard, 2007). By it, college sport performances may have greater societal impact to other aspects of its institutions than thought.

4 METHODS

4.1 *Dataset Description*

The dataset used for this research is an NBA dataset. The data frame is composed by 24 columns and 12804 rows. Each row is one of the players of the NBA players from a specific season mentioned in one of the columns. The dataset comprehends the NBA seasons from 1996 to 2022. It has many continuous features of these different players like measurements, weight and age. As well as many other categorical features as which University team the player started their youth career before being drafted by an NBA team. The round of draft of the NBA. A draft is a round where the worst teams to the best team of the past season are enabled to draft new young players out of college to balance the level of the game and replacing it by the new generations (Hussey, 2021).

Also, other features as nationality, which is one of our most relevant features as the research pretends to explore data analysis regarding nationality bias of the NBA players.

The NBA dataset can be found in Kaggle, a website for Data Science community (NBA Database (kaggle.com)).

On the other hand, there is a self-mined dataset from a source of statistics about the NBA. The website used for this is Basketball-Reference.com. This dataset has been merged to the Kaggle one mentioned previously, in order to identify via dummy coding columns, which player was or was not selected for the NBA All-Star match. Via the Basketball-Reference website a broad amount of important features from the NBA and All-Star matches can be found.

This second dataset was implemented thanks to a Microsoft Excel file mined for this specific research and therefore exported as a pandas data frame to Python VS Code environment.

4.2 *Algorithms and Software*

For the research the selected computer language was Python and the environment used to run all the algorithmic procedures was VS Code (Sanner et al., 1999).

For this classification problem, the objective is to identify by the binary prediction of YES/NO of the selected players for the All-Star team, to what extent there is a nationality bias in the recruitment of these All-Star team players. For this, different models are going to be implemented to selected players, and then control for nationality in the different cases.

The models that were implemented for this parts of the research are binary classification models, one of them probabilistic and the two others tree-based algorithmic models. As it is a binary classification problem of 'Yes' or 'No', or also '1' or '0' respectively, is that these were the output categories.

One of the models was a probabilistic function, which was a Logistic Regression Classifier (hereafter; LR). On the other hand for the tree-based models, the work was done with a Decision Tree Classifier (hereafter; DT), and with a Random Forest Classifier (hereafter; RF).

The logistic regression model is a linear regression model that implements the sigmoid function, and regresses each candidate with a certain number in between 0 and 1 and assigns the output value of the regression to its respective class, hence a 'NO' class if it is closer to a 0 or a 'YES' class if it is closer to a 1, and its meant by 'closer' as it may assign a number in between 0 and 100 in terms for percentage for each instance, and the one were it lies with a bigger probability to is where it will be located, either '1' or '0' or 'Yes' or 'No' respectively. (Solutions, 2016).

A DT is an algorithm capable of been used for either a continuous or a categorical output. For the case of this research a decision tree with a categorical outcome suits best, and more specifically as a binary classification outcome. It works by splitting the dataset with the features that have the highest entropy at the beginning of the iteration and aimed to conclude that iteration measure with the questions with the lowest one, and therefore it produces the outcome class. Entropy is a measurement for uncertainty, which quantifies the disparity of the features in the dataset. Low entropy means that our dataset is very easy to interpret within the different classes, whereas a high entropy makes classification work more hardly separable (Suthaharan & Suthaharan, 2016).

Considering the performance of each of the models it will be possible to identify if the nationality bias of the players is more notorious when more accurate, or if when higher accuracy more bias in the selection of the players. In the case there is a non-existing influence of this feature in the modeling, there can be other features with sociological relevance when having the results of the three different models, Logistic Regression, Decision Tree and Random Forest.

4.3 *Evaluation Method*

The results will be measured with sensitivity, accuracy and specificity for the selection of the NBA All-Star teams. As it is a way of considering how accurate the Yes detection are, by the implementation of sensitivity and how well our algorithms classify the No class with the use of specificity measurement for the three different models, logistic regression classifier, decision tree classifier and random forest classifier.

For the bias implementation analysis in the experiment, the best performing model was choose, and based on it the statistical analysis was conducted through a chi-square test since the categorical variables were the ones of interest. To avoid bias and weight of certain features, the columns like "name" and "country" were extracted from the data set.

The objective was identifying through statistical testing, if by the learning of the algorithm predictions, some group or feature in the dataset may be affecting the prediction of NBA All-Star players. Therefore check if there is statistical significance ($p\text{-value} < 0.05$) showing that the errors in a certain category of the ML model is significant after controlling for a certain variable, if so we would have found some feature or features that can be affecting the 'Yes' or 'No' decision (Tallarida et al., 1987).

4.4 *Preprocessing*

For the master thesis different data mining techniques were necessary to engineer the inputs for the machine learning models. Two datasets for the experiment were merged in order to perform results altogether. A dataset with season performance features and characteristics of the players towards a second data set that was mined for the purpose of this research to extract in a main dataset if each player with each specifics features and characteristics was or was not recruited to be an All-Star season player.

Due to the mentioned process it was possible to calculate algorithmically if the players were selected or not, performing three different binary classification models, due to the dummy columns technique used or the "One hot encoder" code in python programming language.

Before the start of the model runs, an relevant step in the preprocessing of the names features occurred. Before the feature "name" was deleted from the dataset, it needed to be sure that each row had a correct selection with an exact match if selected or not, after running a double iteration in between "names" on each dataset and iterating for the season of each player from 1996 to 2022, the match of the whole dataset was possible. Characters from both datasets need to match identically in order that the model may output the decision class "1" or "0" correctly.

After that, every typo or byte size miss match had been properly cleaned and preprocessed to work with one whole harmonized dataset. For this part, from the package named "fuzzywuzzy" the "fuzz" function was imported. Thanks to this Python library that was installed, it was possible to concatenate a bigger dataset that contains 2551 original NBA names with 187 exact matches from the 188 NBA All-Star names. As for this, the following step was to create dummy columns, or binary classification for the selection or not selected for each season from 1996 to 2022.

This left the data with 25 year rows, so there was one column missing from 2022 until 1996, as there was a lockout for season 1998, for such a reason is that there were no measures available for the NBA All-Star game for that particular year of 1998 (Ringer, [n.d.](#)).

The output of this process is going through a data frame which concatenates the different player features from 1996 to 2022, with the different columns such as the point scoring in their games, assists, games played, percentage of minutes played, their country of origin, names, age, their previous university of competition, their weight, their height, their draft pick and their draft round, and to all this information is that we add the All-Star selection or not as a last column dataset.

Furthermore, each row of this 12844 that represents a specific player are related to an amount of 2551 unique NBA player names, as players participate in the NBA for several seasons. From all this 2551 different names, 187 are a selected group that participated in the NBA All-Star teams at the end of the season, and its what the model was based on.

The aim of this ML models is to have accurate outputs of accuracy with our true labels, in order to check statically to what extent there is a bias, and if some what kind, in the selection of the All-Star players rosters.

For the modelling, the dataset was broke down to a train set with a data amount of 70 percent of it, leaving the latter 30 percent for the validation and test sets. The split was made into 70 for train, 15 for validation and 15 for test given the size of the dataset of 12800 times 25, and since it was not that big a model with bigger training sets is better recommended for a better performing prediction model (Vabalas et al., [2019](#)).

The validation set was carried out through a k-fold equaled 10 for the three different models (Decision Tree Classifier, Random Forest and Logistic Regression), as well as 15 percent of the data set for the test sets of the three models.

Firstly, the training process was done which finds a way in the model to transform the input looking for a specific instance in the dataset which is known by the algorithm, hence accuracy is 100. It works as a mathematics

function in which there is an 'X' input and the function finds an equation to achieve the output 'Y', named as the true values.

This exercise is done several times, as many 'Y' instances in each split of the data. For this research it is 70 percent for training set, so 8991 rows, 15 percent for validation and test sets, so 1927 rows respectively in both cases. The dataset is composed by 12844 rows and 24 columns in total. Given the dimension of it, was that the split was managed for a bigger size in the training to aim for a more robust test set (Vabalas et al., 2019).

In the validation set, with the $k = 10$ folds for the three analyzed models, a grid search was performed to find the best hyper parameters in the validation set, looking to increase the model performance with unseen data for the test set (Shekar & Dagnew, 2019).

The results of the three different models were analyzed using "sklearn metrics" package in python VS Code, utilizing learning performance graphs for the three models, confusion matrices and a feature importance analysis for the best performing model.

Through all this split and grid search it was possible to find a balanced model that performed better in the test set than in the validation set, which was positive news since the model could learn the data in a way that performed better with unseen data. As well, and even after deleting some of the variables like "name" and "country", the results thrown were better than previous literature review papers.

To work with the error analysis, and with it apply statistical testing to see how different groups may affect the errors in the best performing model of the research (random forest), this may infer that after controlling for specific categories from the dataset, the prediction model as well as the people involved themselves may be giving too much relevance to a certain or certain categories.

This made the experiment find insights about the random forest model performance in this data set. This positively impacted the expectations in the research as the features with the most weight in our best performing model were all related to game performance and scoring attributes of the game like points, usage percentage of players, rebounds per game and assists per game.

Although it can also be considered that the usage percentage is already been studied by one of the literature review materials as some racial bias in the selection of players and in minor impact the decision making through out the substitute game play decision which already shows scientific conscious in certain types of bias in the named sports league (Schroffel & Magee, 2012).

Due to the interests of this academic research, was that two column categories were deleted. The aim of the research was to explain the prediction

and finding insights that can unveil the decision process of stakeholders and help from society perspective the decision making of management teams of this institutions. Due to that, was that the categories of "name" and "country" were decided to be left out, as they were considered to have a heavy influence.

In the first place, the variable "Name" can affect heavily the decision making as players like "Michael Jordan" or "Stephen Curry" who had more than 10 selections for this team consecutively, they would be automatically calculated as selected because of the name feature, which in reality is not the case.

For the case of country, most of the players are from the United States but at the same time when foreign players succeed they might have a chance in being part of the prestigious All-Star group. As well and as studied in the literature review, different nations in the NBA league could be associated with a certain race, so would be helping less in the unveiling of a bias for reciting the players. To look into a dataset decreasing strong biases to acquire insights through implementation of data science techniques.

In figure 1 there is a feature importance graphic of the best performing model (random forest classifier) with the selected features for the experiment.

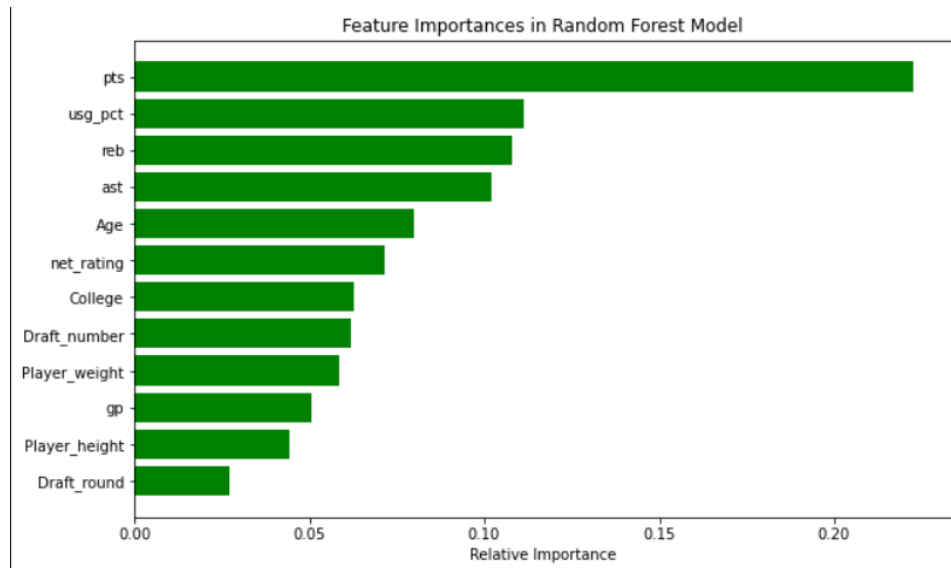


Figure 1: Feature Importance for Random Forest Model

Some of the meaning of the columns names from the dataset where abbreviated to what figure 1 shows. Here the explanation of the abbreviated column names, "pts" refers to points per game in a season, "usg.pct" refers to percentage of game play during a match, "reb" refers to rebounds per

game in a season, "ast" refers to assist provided per game in a season and "gp" refers to the amount of games played in a season.

Many other literature reviews point out the "players popularity", as well as "social media impact" or "press conference characters", "personality types", etc. This named circumstances can be considered as important weight variables but unfortunately were not measured precisely. Because of the mentioned situation, this can be considered a limitation in the research process, and something to be approached by other researchers in similar fields for future experiments.

This can explain some of the accuracy gap from 0.93 accuracy in our test set to even higher values, as well as the 0.68 for specificity in our best ML model, and even increase the 0.98 performance for the sensitivity measure, giving a total macro average of 0.83.

Moreover, the analysis of this master thesis consisted in leaving some of the features aside and then elaborate in the results of the best model (random forest) and analyze if any categories can be inclining the balance to a side in the decision of the people involved.

4.5 *Models and algorithms used*

The prediction outcome was a binary classification, meaning that the outputs of the model were either number 1 answering to a positive instance "selected", and number 0 answering to a negative instance of "not selected".

As in the purposes of answering the third research question, one of the objectives was to work with model comparison performance. Due to that, and our classification problem of player selection, is that the three ML models that were chosen were the following, Decision Tree Classifier (hereafter; DT), a Random Forest Classifier (hereafter; RF), and a Logistic Regression Classifier (hereafter; LR).

The DT is a ML model that does its computational findings through yes or no statements, dividing in each step of the tree or each node with statement with the lowest entropy to the highest entropy. Entropy is a coefficient between 0 and 1 which shows how different our outputs respect to a certain feature are, so decision tree classifiers start the process of a prediction output by splitting the data from more general instances, with less division of the data, and aiming to the next node into the variable with the following highest entropy, and so on. The process ceases when the feature with the highest level of entropy stands. This instance will be the one with the more random outputs.

Also a LR classifier was implemented. A logistic regression classifier model has other particularities worth pointing out. This model works as a linear regression, but outputs categories and a brief explanation of

how that happens will introduce it. When working in linear regression models, different from logistic regression, variables have basically a certain ratio and given the input of each of the variables to that ratio, is that a function can regress the continuous output for a certain instance in a linear regression model. Reminding once again that a linear regression works for continuous values, hence numerical output that may not have theoretically speaking any intermediate values between two outputs.

The LR model required some other steps in the feature engineering given the algorithm processing. For the data preparation, besides handling missing values which have already been fixed in previous steps, it is important to handle an encoding of the categorical variables as well as normally distributing the continuous variable values for this specific Logistic Regression model. To deploy the model calculations, the “Logistic Regression” function from “Scikit-Learn” package in Python environment was imported.

A relevant outlook in the LR model and that it differentiates out by the previous RF and DT classifiers is the error function of the algorithm. The error function in LR is the Log loss function or logarithmic loss, it quantifies the difference from predicted to observed values, and its discrepancy for the other tree-based models in that its classification lies in a probabilistic measurement, rather than in an entropy split classification.

For this calculation, also the correction or penalization of the model itself varies, as the wrong classifications will be heavily penalized aiming to neutralized the mistakes to produce a bigger number of true outcomes.

Given the different mathematical natures of these models, it is that they behave and translate the errors of the models with different expressions. Another specific difference lies when the algorithm produces an error, the penalization is bigger as it needs to adjust its parameters to iterate more precisely for future instances. Whereas the decision tree and the random forest, both tree-based methods, process data to divide it in homogeneous splits working with the concept of entropy and division in its features, and aiming to classify the category outcomes logically.

4.6 *Evaluation*

To answer the three research questions the machine learning models comparison in the prediction of the selected players for the NBA All-Star matches were studied together with chi-square statistical tests results.

For its answering of the research question is that two ML approaches were selected, a probabilistic with a logistic regression classifier and two different tree-based methods a random forest classifier and a decision tree classifier.

To initialize the running of models with the LR model, the following libraries and packages from python were needed. The main libraries were "sklear.linear.model" to import the Logistic Regression function as well as the "sklearn.preprocessing" library to import the "standard scaler" and "label encoder" to preprocess the data. are the "sklearn.preprocessing" where "StandardScaler" and "LabelEncoder" . This functions worked with the data transformation to normalize and encode different many of the columns in the data set, enabling the model run in its probabilistic nature. Big values may affect the calculations when working with logarithmic functions, therefore all this preprocessing is needed for a probabilistic model.

The columns that were normalized in the data set for the LR model were six, player weight, player height, number of games played per season, assists per season, net rating in a season, points per game in that season, as well as their draft number (position in the draft), and number of rebounds per game in that season.

The draft feature is the hiring process for NBA basketball players, in where teams pick the newcomers or more socially known as "rookies" as well as the international players of the NBA season and in other sports leagues alike in the US. New players and internationals players also are hired by the NBA franchise and are paid to do a preseason and then a process of selection takes place. It is a system of rounds, in where the worst ranked team of the previous season gets the first draft pick and so on using the last seasons positions ranking in order to balance the future in between the best and the worst teams of the league, and in the NBA there are just two rounds for the draft pick (Hussey, 2021)o.

On the other hand, the dataset had other categorical values such as draft number, college, and draft round. Country indicates the nationality of the player. As previously mentioned features such as name and country were took out to extract insights from a dataset without a nationality or name influence, which remained the research with satisfactory results.

College indicates which educational institution they played in previous to become professional NBA players, and draft rounds involve only two rounds, meaning there are two turns to select the newcomers after or previous to the start of a new season. For these categorical variables is that the features were encoded, differentiating to the continuous variables which were statistically normalized.

After the feature engineering for the data processing in the LR model, the performance of the algorithm was of 0.90 for the best cross-validation score and for the test set. The sensitivity performance of the model situated in 0.97 for the true negatives or 'o' category instances in (not) selected players, and the sensitivity for positive instances or '1' was situated in 0.53.

Respectively, the weighted average of this ML model was of 0.90 and the macro average a 0.75, in the measurements of our interest (sensitivity and specificity).

The performance of the Logistic Regression model in the train and validation sets and a confusion matrix of its results were calculated. Every confusion matrix in the research were represented using a color blind tolerant plot display.

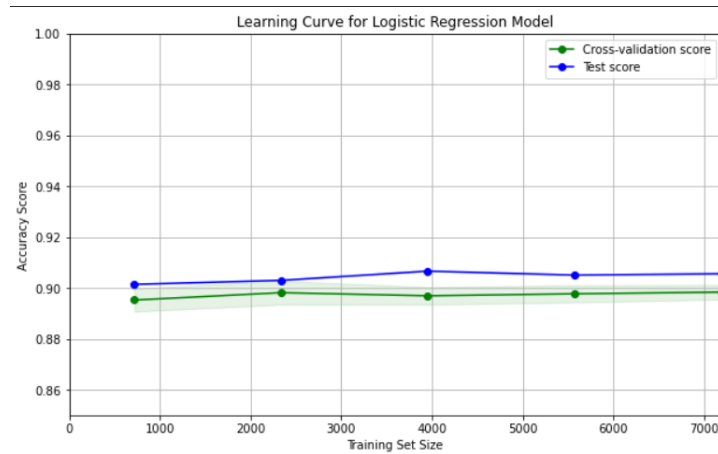


Figure 2: Logistic regression validation and test sets performance



Figure 3: Confusion matrix performance for the logistic regression model

Secondly, another two models were conducted to discuss the model performance in between different models. On the one hand, a Decision Tree Classifier, imported from the 'sklearn.tree' package in Python which provided an accuracy of 0.89 for the test set, with a sensitivity of 0.93 for

the negative or '0' instances and a 0.64 for the positives or '1' instances. The macro average for the model was 0.77, and the weighted average for the DT was 0.89.

Hereby figure showed the results of the different model sets as well as another image with a confusion matrix of its results.

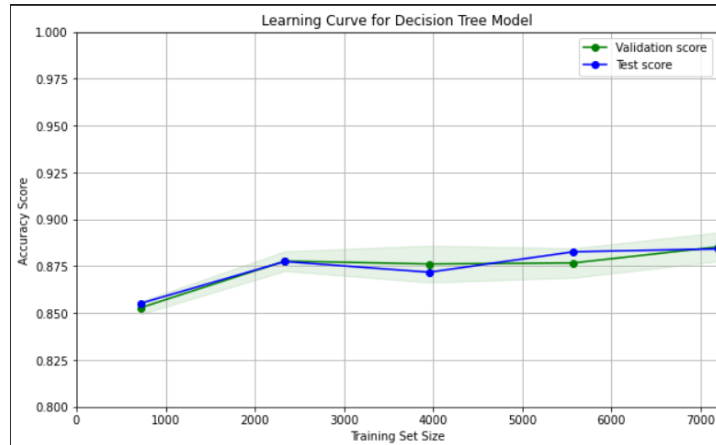


Figure 4: Decision tree validation and test sets

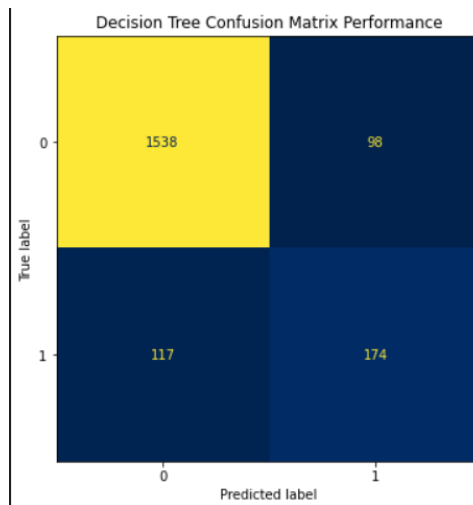


Figure 5: Decision tree confusion matrix

Last but not least, the random forest classifier was ran. It was implemented by the 'sklearn.ensemble' package which enables the function Random Forest Classifier, and 'sklearn.model selection' to import a Grid Search and a k-fold ($k = 10$) for the validation set. The two tree-based models are such because of the nature of its calculation, a RF performs

what a DT calculates multiple times, reducing the erroneous instances for a more generalized and balanced model.

After the feature engineering, the Grid search and a K-fold ($k = 10$) the best model performance was found in the random forest classifier. The accuracy of the RF presents an average of 0.93 in its predictions, and a performance of 0.98 for the negative or '0' class instances, and of 0.68 for the positive or '1' class instances. Therefore, the weighted average and macro average presented its best performances with a 0.93 and a 0.83 respectively.

Hereby, the learning graph for the validation and test sets and a confusion matrix for the results of the random forest classifier.

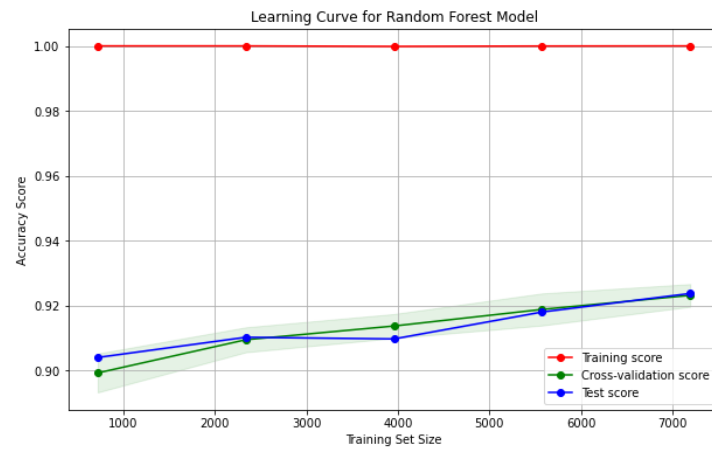


Figure 6: Random forest validation and test sets performance

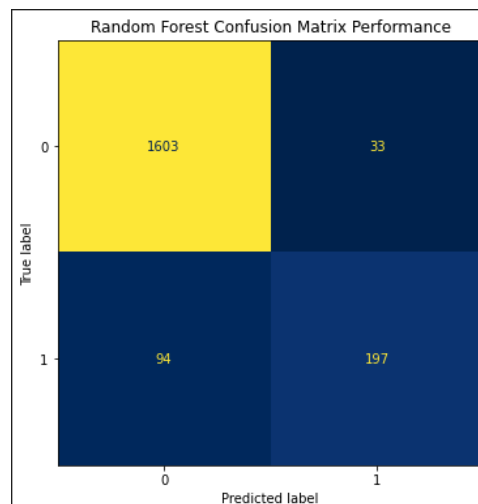


Figure 7: Random forest confusion matrix

After the mining and concatenating the two data sets, cleaning all its instances, taking features like country and names to reduce bias, implementing the output 'YES/NO' column, and implementing the three different ML models using a grid searches for the validation sets to get best tests accuracy, the best performing model for the study using NBA players data from 1996 to 2022 All-star players was shown by the random forest classifier.

It is relevant to point out that the data set used was intended to avoid bias while trying to execute the predictions. Applying less categories related to features like race and direct preference like name, to manifest a more conscious decision making from the stakeholders.

The statistical test used was the chi-square, since the categories were considered a variable that may been having more weight in the selection process than it ma actually matter. For this, three categories were remaining in the data set, "draft round", "draft number", and "college".

The statistical test were conducted over the dataset with the true positive values in the random forest classifier model. This model was the best performing model, so selected to continue with the experiment (accuracy = 0.93), and the highest flaw in the calculations were found in the true positive instances with 0.68.

The chi-square tests were performed in the python vs code environment, from the "scipy.stats" library. The first performance of the test was in the draft round and the p-value for it was 0.234. Secondly the draft number chi-square test was performed and the p-value obtained was of 0.109. After that, the chi-square for college category was performed and a significant p-value equaled to 0.011 was the result.

There is a 0.011 p-value when testing for chi-square statistical test weather in true positive against false positives, as well as true negatives against false negatives in the college group samples. This may be interpreted as bias since a variable is receiving bigger weight than it should, since there is a correlation and a significant p-value that the different in the selection of players is been affected by this category.

In different words, and related to what was previously in the literature review, different players may receive a bigger hype from the relevant stakeholders and are expected by the model to be selected, conditioned by the college they attended in the early stages of their basketball careers. This is very interesting considering other research papers related to similar aspects suggested that tuition fees, application number and rejection ratio may all increase together when a college performs better for college football (Stinson & Howard, 2007).

5 RESULTS

In the following table it is possible to visualize for the three different models proposed for the third research questions as well as answering the first question if a machine learning model could predict such findings.

The table shows the most relevant measurements used in this research which were accuracy, true positives and true negatives percentages and the macro average.

The random forest classifier over performs the rest of the models, considering the most relevant measure to answer research question three, based on its accuracy. The model that showed the best results after the RF was the logistic regression and then followed by the decision tree. In Table 1 a comparison of the results obtained in the experiment.

Table 1: Model performance comparison

Model	Accuracy	True Positives	True Negatives	Macro Average
Logistic Regression	0.90	0.53	0.90	0.75
Decision Tree	0.89	0.60	0.94	0.77
Random Forest	0.93	0.68	0.98	0.83

As for the second research question, the following table showed the results of the chi-square tests. The variable college was identified as the only one with a significant p-value score for the three different categories analyzed in the true positive instance data set for the best performing model, the random forest classifier (from research question three). Hereby in Table 2 the results.

Table 2: Chi-square tests results for categories in true positives of the random forest model

Category	P-value
Draft Round	0.234
Draft number	0.109
College	0.011

6 DISCUSSION

The second research question of this Master thesis was to identify if there was any bias for the selection of players, or any relevant insight in order to help sport management teams, sport data driven departments, mainstream

media channels, the players, or either fans themselves. For this regard, the feature importance was relevant in the fact that the biggest weight variables were those ones strictly related with the player performance or players scoring for a season, like points, player usage percentage, rebounds and assists, representing around the 0.50 of the model calculations. Producing a model with less categories that may affect heavily on the bias like "name" and "country", the variables "college" came somehow lower in the RF split as it was at the ending of the feature importance graph.

On the other hand, the RF performed more than well for negative or '0' class instances, with a result of 0.98 for this outcome. However, the accuracy score drops when calculating the positive instances or '1' predictions to a much lower 0.68. So is that the next step of the experiment was focused on testing statistically in the positive instances to check if there were variables that have been indeed affecting this decision in players careers.

By this, the research tries to reveal this selection of players from a conscious perspective, unveiling decision making by these stakeholders like the specialized media, NBA coaches and fans together with club management who may elaborate better decisions when considering scouting.

As mentioned in the methods section and as it was practiced, the focus remained in the sensitivity (true positives) and the specificity (true negatives). Consecutively, the model performed excellent in the specificity aspect, but showed some flaws in the sensitivity prediction process.

Given the best performance, and answered the first research question of which algorithm can identify this task more accurately, is that the random forest results were analyzed into further detail, and continuously as the research has to be conducted, the results were tested statistically to explain insights from a sociological perspective.

Besides, other studies had satisfactory results (92.5 accuracy) measuring a similar problem regarding research question one, about predicting through machine learning, it is interesting to consider how this results may continue to be improved due to some research limitations in the master thesis.

These limitations could be to work on a bigger data set for future researches, as well as consider other relevant topics like "personality", "popularity", "interactions with the media", using different measures for it like social media platform followers and amount of press interviews per season.

In the literature review section of this academic research there were different bias regarding sports in different disciplines of study, socio-economic, racial, and how college sports may affect rating improvements in different universities from the United States.

The aftermath of these thesis results were that a player who went to a certain college has a better shot on becoming very successful than the others, but for future researchers the question to analyze could be if it is a problem that is continuously fed by the same issue or not. Is because some players that come from certain institutions receive more money from other related stakeholders, or should a mechanism to neutralize this bias in success of youngster players be conducted to equalize the opportunity outcome in the future of athletes. It is indeed an institutional approach.

As well, another interesting remark found by another researcher was that in the league of study, the NBA, managers with a certain ethnicity are prone to consider better certain player of an ethnicity alike their current coach himself, shown with a bigger correlation at the moment of the team assembling previous to the game, and in lower measurement while taking decisions in the process of the game like performing player changes during the game from the technical directors.

Agreeing with the last statement and with other ones such as the improvement of college requests and funding in different universities due to winning streaks in American football, is that also question if there is a societal flaw in the conscious sense that sports in universities and university budget imbalance may be study from a more fundamental approach and if that is making society take action in the outcomes of this professionals, either in sports or other domains.

The experiment threw quantitative evidence in the $p\text{-value} = 0.011$ of the chi-square test of the true positive results (0.68) in the random forest model (0.93 accuracy) after controlling for the college category in the dataset. The most important remark of the academic research, besides proving a point for this specific discipline or even sports as a whole, is that the same methodology may be extended for other processes in society decision making such as buying a house, receiving a loan, getting hired for certain jobs, or getting accepted by a certain university.

The college variables was shown to be the only category to present a bias in the decision of the algorithm, from the three ordinal categories analyzed, college, draft rounds and draft number (Hildebrand et al., 1977).

The chi square test takes different categories and compares in between groups to see if there is a significant effect, $p\text{-value}$ of under 0.05, in the effect of a certain group on between categories compared to the rest in the results of the studies. If the $p\text{-value}$ is bigger than 0.05 means that there is no significant effect also understood like the outcomes for those variables are just randomly given in a specific way, but if the $p\text{-value}$ is inferior to 0.05 the interpretation changes to a significant effect meaning that it is not randomizing the result of the experiment respect to a certain categorical variable.

The chi square statistical test was indeed conducted for these three categories, obtaining the following p-values. After controlling for the category "draft round" the p-value equaled 0.234 hence non-significant. After controlling for category "draft number" the p-value equaled 0.109, so non-significant as well. Whereas for the category 'college' the p-value equaled 0.011 meaning there is statistical significance with the experiment of the performance of positive instances for a random forest, and this variable "college".

Realising this, the feature college is receiving a weight such that can decide either in the selection or non selection of these players. As an institutional approach and more inside academic research another study could be conducted explaining how this universities receive more sports success, and controlling for different variables such as geography, students features, funding of the institutions.

This meaning that college history from athletes might be receiving a bigger relevance than it actually could have for such a group, if we considered a bias exists. If the decision making can be studied outside the biased, it may lead to an information gain to develop other plans.

This can be correlated to previous literature review of socioeconomic background, or even certain bias or statistical sensitive affinity from the ones who make the decisions. Instead of, in this case, the responsible being the manager, for this All-Star selection the cause would be fan ballot, players and the selected media specialists who all together choose the NBA All-Star players.

As an aftermath, it can be interpreted that in the world of NBA stars, the college history of players it is influencing the perspective that, coming from a certain institution as a former athlete might incline the balance to the All-Star path or the non-All-Star selection path.

For further investigations it would also be interesting to find out, to what extent are competition college affecting career performances in long term, and also to what extent career opportunities, as in a personal view the career prospect of this individuals from different colleges might have an overrated weight, as the level of competition in National Competition American Association (college level sports) is already considered as professional competitiveness, only having aside the economic aspect of world sports retribution (Sandy et al., 2004).

7 CONCLUSION

The experiment conducted showed a bias in the selection of players due to a college history of students previous to their professional career. Due to the numbers threw in the analysis it is quantitatively confident to assure

that a bias in the decision making due to the variable college exists in the selection. For other studies, and relating to the socioeconomic backgrounds of students in college, it may be interesting for future researchers to investigate to what extent this difference in colleges athletes that may lead to more successfully careers was already settled or influenced by other specifics in athletes that may shorten their careers outcome.

From a different perspective, in other mainstreams sports such as football, many players become super stars coming from humble backgrounds, meaning that youth background and beginning in these disciplines may not have a big participation in what the player will develop to be transformed into, and from society prospect this is very contributing to the discussion but might be more insightful if conducted by other studies into further details and analysis. With this been mentioned, is also relevant to consider the identification of bias also in sports controlling for different aspects of the individuals taking part. This can also be adapted to many society domains, which can improve our decision making as thanks to machine learning and statistics we can support this findings from a quantitative perspective.

Numerous studies have demonstrated the capability of Machine Learning models, with lower performing accuracy's, to predict NBA All-Star rosters, and in this research well performing predictions were possible to appreciate. Previous researchers in similar areas of study, had also revealed biases like racial and socioeconomic influences for sport making decisions, and sport success careers. This evidence supports the feasibility of exploring decision biases in sports through the field of data science. Ultimately, investigating such biases is crucial and applicable not only to the NBA and other sports but to various societal domains as well.

REFERENCES

- Gandelman, N. (2009). Selection biases in sports markets. *Journal of Sports Economics*, 10(5), 502–521.
- Grossman, N. (2014). What is the nba. *Marq. Sports L. Rev.*, 25, 101.
- Guevara, J., Martín, E., & Arcas, M. (2021). Financial sustainability and earnings management in the spanish sports federations: A multi-theoretical approach. *sustainability* 2021, 13, 2099.
- Hildebrand, D. K., Laing, J. D., & Rosenthal, H. (1977). *Analysis of ordinal data*. Sage.
- Hussey, A. (2021). *Nba draft analysis* [Doctoral dissertation, Dublin, National College of Ireland].
- Langel, M., & Tillé, Y. (2011). Statistical inference for the quintile share ratio. *Journal of Statistical Planning and Inference*, 141(8), 2976–2985.

- Levine, G. R. (2019). *All-nba team voting patterns: Using classification models to identify how and why players are nominated* [Doctoral dissertation, Ohio University].
- Ringer, T. (n.d.). *Lockout seasons 1999* [citation of source the ringer].
- Sandy, R., Sloane, P. J., Rosentraub, M. S., Sandy, R., Sloane, P. J., & Rosentraub, M. S. (2004). College sports in the usa and the role of the ncaa. *The Economics of Sport: An International Perspective*, 257–284.
- Sanner, M. F., et al. (1999). Python: A programming language for software integration and development. *J Mol Graph Model*, 17(1), 57–61.
- Schroffel, J. L., & Magee, C. S. (2012). Own-race bias among nba coaches. *Journal of Sports Economics*, 13(2), 130–151.
- Shekar, B., & Dagneu, G. (2019). Grid search-based hyperparameter tuning and classification of microarray cancer data. *2019 second international conference on advanced computational and communication paradigms (ICACCP)*, 1–8.
- Soliman, G., Misbah, A., Eldawlatly, S., et al. (2017). Predicting all star player in the national basketball association using random forest. *2017 Intelligent Systems Conference (IntelliSys)*, 706–713.
- Solutions, S. (2016). What is logistic regression. *Retrieved from*.
- Stinson, J. L., & Howard, D. R. (2007). Athletic success and private giving to athletic and academic programs at ncaa institutions. *Journal of Sport Management*, 21(2), 235–264.
- Suthaharan, S., & Suthaharan, S. (2016). Decision tree learning. *Machine Learning Models and Algorithms for Big Data Classification: Thinking with Examples for Effective Learning*, 237–269.
- Tallarida, R. J., Murray, R. B., Tallarida, R. J., & Murray, R. B. (1987). Chi-square test. *Manual of pharmacologic calculations: with computer programs*, 140–142.
- Vabalas, A., Gowen, E., Poliakoff, E., & Casson, A. J. (2019). Machine learning algorithm validation with a limited sample size. *PloS one*, 14(11), e0224365.
- Wang, J., & Fan, Q. (2021). Application of machine learning on nba data sets. *Journal of Physics: Conference Series*, 1802(3), 032036.
- Zhang, L. (2017). A fair game? racial bias and repeated interaction between nba coaches and players. *Administrative Science Quarterly*, 62(4), 603–625.